

Joey Yizhi Li

📍 Toronto

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🔗 [Joey-the-Creater](#)

Education

University of Toronto

Sept 2024 – May 2029

BASc in Engineering Science (Major in Machine Intelligence)

- **cGPA:** 3.93/4.0, Dean's Honours List (3×)
- **Relevant Coursework:** Deep Learning Fundamentals, Algorithms & Data Structures, Digital & Computer Systems, Electrical Circuits, Structures & Materials, Vector Calculus, Fluid Mechanics

Research and Projects

Commercial Aircraft Variant Recognition System

May 2025 – Aug 2025

- Designed a fine-grained aircraft recognition system using a **ResNet-50 CNN** with a **Spatial Pyramid Pooling (SPP)** layer to classify **59 commercial aircraft variants** from images of arbitrary resolution.
- Achieved **65.08% validation accuracy** at the variant level, substantially outperforming a Bag-of-Visual-Words baseline.
- Collected and annotated **15 new aircraft variants** and implemented a **7-factor data augmentation pipeline** to improve robustness and generalization.
- **Tools:** PyTorch, NumPy, OpenCV, scikit-learn

LeSort: Automated LEGO Sorting Machine | [GitHub](#)

Feb 2025 – Apr 2025

- Built an automated cyber-physical system to sort LEGO pieces by **color, size, and type** using computer vision and embedded control.
- Integrated a configurable CVML model with a real-time GUI feedback system for adaptive sorting strategies.
- Achieved **90% sorting accuracy** with an average processing time of **3.5 seconds per piece**.
- **Tools:** OpenCV, Tkinter, Arduino, CAD

ChaGratation: LLM-Based Mobile App Test Migration | [GitHub](#)

Jan 2023 – Dec 2023

- Developed an LLM-driven framework for **automated mobile app test migration** in the AXiS research program (Tencent).
- Leveraged **prompt engineering** to perform semantic matching between UI elements via inferred UI Transition Graphs.
- Achieved **90% migration accuracy** with an average runtime of **3 minutes**, outperforming prior rule-based approaches.
- Supervised by Prof. Tao Xie (PKU) and PhD researcher Dezhi Ran.
- **Tools:** OpenAI's API client, Android Emulator, Appium 2.0, ADB, Python

Experience

Summer Research Intern, Fudan University

May 2025 – Sept 2025

- Conducted research on **RAG-enhanced LLM autonomous agents** for software test execution under Prof. Dong Zhen.
- Abstracted mobile UI Transition Graphs (UTGs) using **Chain-of-Thought prompting** to improve test coverage and reasoning.

Honors

University of Toronto In-Course Scholarship

Aug 2025

- Awarded **\$2,997** for outstanding academic performance

Canadian Mathematical Olympiad (CMO) Participant

2023 – 2024

- Scored **71/80** in the Canadian Open Mathematics Challenge; ranked top 50 nationally

USA Mathematical Olympiad (USAMO) Qualifier

2022 – 2023

- AMC 12 Distinguished Honor Roll (Top 1%), AIME: 12/15

HiMCM Finalist (Top 6%)

Nov 2021

- Co-authored a 20-page modeling paper on optimal off-grid battery systems

Technical Skills

Programming: Python, Java, C/C++, MATLAB, JavaScript, HTML/CSS, SystemVerilog, RISC-V, R

Tools & Platforms: PyTorch, Linux, Git, Android Studio, Appium, FPGA, ModelSim, CAD